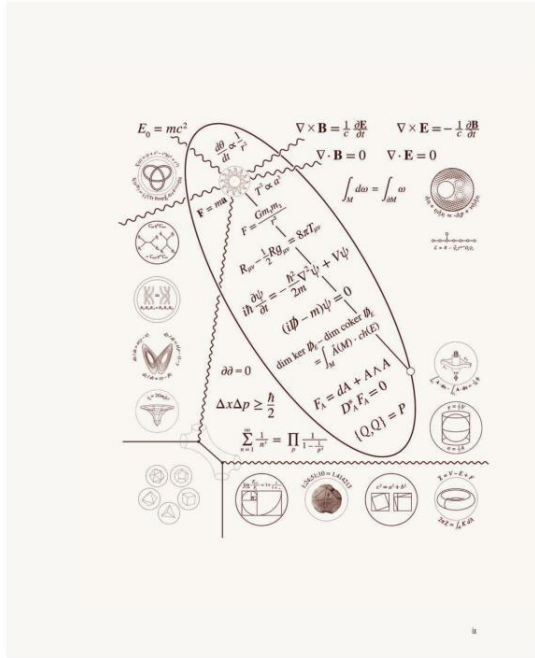


Iconic Wall Change Log

Original Simons Center wall reference → final Mister Consistency fabrication artwork

Revision 2 · July 9, 2026 — Maxwell's equations recast in differential forms; Bekenstein–Hawking entropy added (background J)

Original reference diagram (catalogue)



Final vector fabrication artwork (rev. 2)

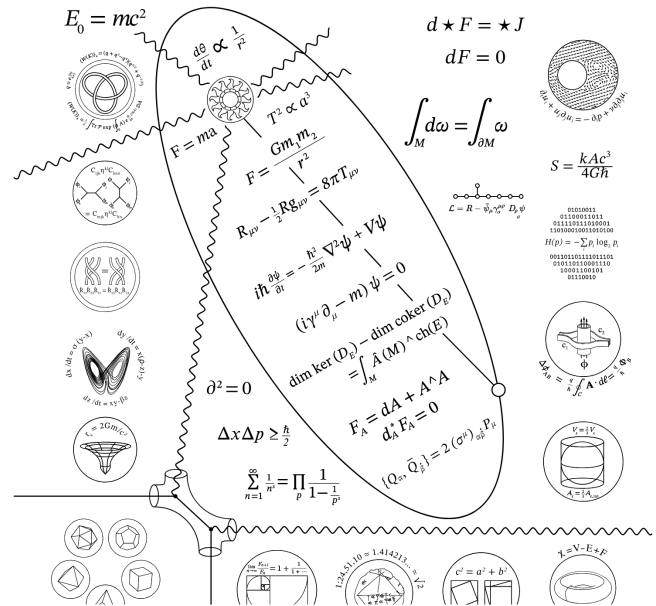


Figure 1. Original catalogue wall diagram compared with the revised 42×42 vector fabrication artwork (July 2026, revision 2).

1. Executive summary

- The final Iconic Wall remains a faithful reconstruction of the original Simons Center wall: the central ellipse, peripheral medallions, background equations, and overall historical math/physics progression were preserved.
- The dominant change is production-oriented: the original stone/relief/photo catalogue reference was converted into monochrome, fabrication-ready vector line art for laser-etched or sandblasted steel panels.
- The final package adds two new background entries: Shannon's information theory, shown as binary strings and the entropy equation $H(p) = -\sum p_i \log_2 p_i$, and the Bekenstein–Hawking black-hole entropy $S = kAc^3/4G\hbar$. This expands the background key from A-H to A-J.
- Maxwell's equations were recast: the four vector-calculus equations ($\nabla \times \mathbf{B}$, $\nabla \times \mathbf{E}$, $\nabla \cdot \mathbf{B}$, $\nabla \cdot \mathbf{E}$) were replaced by the differential-forms presentation $dF = 0$ and $d\star F = \star J$, matching the geometric language used elsewhere on the wall (Stokes, Yang–Mills, $\partial^2 = 0$).
- The legend was rewritten for consistency, readability, and specificity: capitalization was normalized, long entries were shortened, image credits tied to removed raster sources were dropped, and several labels were made more precise.
- Several formulas were cleaned or clarified, most notably Dirac, Yang–Mills, supersymmetry, Archimedes, boundary-of-boundary notation, and the mass-energy label.

2. Major changes by category

Area	Original	Final	Change made
Overall wall	Carved/sculpted wall documentation and catalogue diagram.	Rebuilt as clean, square, fabrication-ready vector line art.	Major production change: from photographic/relief reference to flat black vector art.
Overall format	Original installation is tall: 279 × 240 × 2 inches; catalogue describes a 23 ft × 20 ft wall.	Final fabrication outputs are square 42 × 42 inch and 32 × 32 inch panels, plus a 10 × 10 inch legend plaque.	Scale and aspect ratio changed for steel-panel fabrication.
Color/material look	Stone relief, shadows, subtle tonal texture, catalogue teal headings.	Pure black line art on no background rectangle; no gradients, raster images, or shadows.	All tonal/material cues removed for engraving/etching clarity.
All medallions	Mix of diagram line work, relief, and several photo/image-based medallions.	All medallions redrawn as consistent vector strokes.	Unifies style and improves reproducibility at reduced sizes.
Central ellipse	Large diagonal ellipse with sun/focus, planet/path, equations 1–10.	Retained; line weights, spacing, and surrounding formulas cleaned for square layout.	Core composition preserved.
Wavy photon/Feynman lines	Sculptural relief/photo-like wavy lines crossing wall.	Converted to high-contrast vector stroke pattern.	Fabrication readability.
Background equations	Four vector-calculus Maxwell equations; background items A–H.	Maxwell recast as $dF = 0$ and $d \star F = \star J$; Shannon (I) and Bekenstein–Hawking (J) blocks added.	Notation modernized; two new background entries.
Legend system	Original key uses I–XIII, 1–10, and background A–H.	Final key uses I–XIII, 1–10, and background A–J.	Shannon and Bekenstein–Hawking additions required two new background key entries.

3. Legend changes: original → final

The final legend plaque is not a verbatim copy of the catalogue legend. It is a rewritten key for a new fabrication object. The core numbering system remains, but the wording is normalized and two new background items are added.

Key	Original wording	Final wording	Change
Heading	Key to Equations and Diagrams	Key to equations and diagrams (I–XIII)	Lowercase style normalized; numbering range made explicit.
I	The Jones Polynomial	The Jones polynomial	Capitalization normalized.
II	Associativity in Quantum Field Theory	Associativity in quantum field theory	Capitalization normalized.
III	The Yang-Baxter Equation	The Yang-Baxter equation	Capitalization normalized; dash treatment standardized.
IV	The Lorenz Attractor (Image: Scott Sutherland)	The Lorenz attractor	Capitalization normalized; original image credit removed because the final is redrawn line art.
V	The Schwarzschild Black Hole	The Schwarzschild black hole	Capitalization normalized; “Schwarzschild” spelling retained/verified.
VI	The Five Platonic Solids	The five Platonic solids	Capitalization normalized.
VII	The Golden Ratio	The golden ratio	Capitalization normalized.
VIII	The Babylonian tablet YBC 7289 (Image: Bill Casselman)	The Babylonian tablet YBC 7289	Original raster/photo attribution removed from the final legend; final tablet is vector line art.
IX	The Pythagorean Theorem	The Pythagorean theorem	Capitalization normalized.
X	The Gauss-Bonnet Theorem	The Gauss-Bonnet theorem	Capitalization normalized; hyphen retained.

Key	Original wording	Final wording	Change
XI	Archimedes' calculation of the volume and area of the sphere	Archimedes' sphere and cylinder	Long descriptive label shortened to the visual/subject focus of the medallion.
XII	The Aharonov-Bohm Effect	The Aharonov-Bohm effect	Capitalization normalized; hyphen retained.
XIII	The Navier-Stokes Equations (Image: J.D. Kim, AMS, Stony Brook)	The Navier-Stokes equations	Capitalization normalized; original image credit removed because the final is redrawn line art.
Ellipse heading	In the Ellipse	Within the ellipse (1-10)	Heading made more natural and numbering range made explicit.
Ellipse note	(Kepler's 1st law represented by star, ellipse, planet)	(Kepler's 1st law represented by star, ellipse, planet)	Kept conceptually, with surrounding legend typography cleaned.
2	Newton's Second Law of Motion	Newton's 2nd law of motion	Numeral style harmonized with Kepler's 1st/2nd/3rd laws.
Background heading	On the Background	On the background (A-J)	Capitalization normalized; expanded from A-H to A-J.
A	$E=mc^2$	Einstein's rest mass-energy equivalence	Formula-only label replaced with clearer descriptive title.
D	Supergravity	Minimal supergravity Lagrangian	Made specific to the actual displayed formula.
H	Feynman Diagrams	Feynman and string interaction "pair-of-pants"	Expanded to reflect the combined particle/string interaction drawing.
I	No original entry	Shannon's information theory	New background item added to the final artwork and legend.
J	No original entry	The Bekenstein-Hawking entropy formula	New background item added in the July 2026 revision.

4. Element-by-element content, notation, and drawing changes

The following tables list every named item in the original key, the ten ellipse items, every original background item, and the new final background items. "Retained" does not mean visually identical; in nearly every case the final version was redrawn or retypeset as consistent black vector art.

4.1 Peripheral diagram and medallion items (I-XIII)

Item	Original reference	Final version	Change type
I Jones polynomial	Trefoil knot with Jones polynomial/Chern-Simons expression.	Retained and redrawn as vector line art; legend capitalization normalized.	Redraw/typography.
II Associativity in quantum field theory	Tree-level associativity diagram in medallion.	Retained and redrawn; legend capitalization normalized.	Redraw/typography.
III Yang-Baxter equation	Braiding equality diagram with R-matrix expression.	Retained and redrawn; label capitalization normalized.	Redraw/typography.
IV Lorenz attractor	Lorenz butterfly image credited to Scott Sutherland.	Redrawn as vector attractor with surrounding equations; image credit removed from legend.	Photo/image-to-vector change.
V Schwarzschild black hole	Black-hole embedding diagram medallion.	Retained and redrawn with clean vector curves; spelling verified.	Redraw/spelling check.
VI Five Platonic solids	Five separate solid diagrams.	Retained and redrawn with consistent outline circles and strokes.	Redraw.
VII Golden ratio	Golden rectangle/spiral medallion with limit/continued fraction.	Retained and redrawn; label capitalization normalized.	Redraw/typography.

Item	Original reference	Final version	Change type
VIII Babylonian tablet YBC 7289	Catalogue uses a tablet photograph/image credited to Bill Casselman.	Converted to simplified line-art tablet medallion; original image credit removed from legend.	Photo-to-vector change.
IX Pythagorean theorem	Wordless proof diagrams in medallion.	Retained and redrawn; label capitalization normalized.	Redraw/typography.
X Gauss–Bonnet theorem	Torus medallion and formula.	Retained and redrawn with cleaner torus and formula placement.	Redraw.
XI Archimedes sphere/cylinder	Label: calculation of the volume and area of the sphere; formula shown generically.	Shortened legend label; formula clarified as sphere/cylinder relationship, e.g. $V_s = 2/3 V_c$ and $A_s = 2/3 A_{c,tot}$.	Label + notation clarification.
XII Aharonov–Bohm effect	Magnetic flux tube/path diagram.	Retained and redrawn; formula and paths cleaned.	Redraw/typography.
XIII Navier–Stokes equations	Flow image credited to J.D. Kim/AMS/Stony Brook.	Vectorized as line-art vortex/flow diagram; image credit removed from legend; medallion repositioned to the upper right in the revised layout.	Image-to-vector change.

4.2 Ellipse items (Kepler + 1-10)

Item	Original reference	Final version	Change type
Kepler’s 1st law	Star/sun, ellipse, planet represented compositionally.	Retained; ellipse and sun lines cleaned in square vector layout.	Redraw/layout.
1 Kepler’s 2nd law	$d\theta/dt \propto 1/r^2$.	Retained; retypeset and repositioned in ellipse.	Typographic cleanup.
2 Newton’s 2nd law	$F = ma$.	Retained; legend changed “Second” to “2nd” to match nearby entries.	Legend style.
3 Kepler’s 3rd law	$T^2 \propto a^3$.	Retained; retypeset/repositioned.	Typographic cleanup.
4 Newton’s gravitation	$F = Gm_1m_2/r^2$.	Retained; retypeset/repositioned.	Typographic cleanup.
5 General relativity	Einstein field equation in ellipse.	Retained; retypeset with cleaner spacing.	Typographic cleanup.
6 Schrödinger equation	Nonrelativistic Schrödinger equation.	Retained; retypeset with clearer h-bar, gradient/Laplacian, and potential notation.	Typographic cleanup.
7 Dirac equation	$(i\partial - m)\psi = 0$ in the original catalogue diagram.	$(i\gamma^\mu \partial_\mu - m)\psi = 0$ in final artwork.	Notation changed from compact/slashed form to explicit gamma/partial form.
8 Atiyah–Singer index theorem	Index formula in ellipse.	Retained; retypeset/repositioned.	Typographic cleanup.
9 Yang–Mills equations	$F = dA + A \wedge A$ and $D^*F = 0$ style notation.	$F_A = dA + A \wedge A$ and $d_A^*F_A = 0$ style notation.	Subscripts/operator notation clarified.
10 Supersymmetry	$\{Q,Q\}=P$ in original legend/catalogue treatment.	Expanded spinor-indexed anticommutator relating supercharges to momentum.	Schematic formula made more explicit.

4.3 Background items (A–J)

Item	Original reference	Final version	Change type
A Mass–energy equivalence	Background label: $E=mc^2$.	Legend becomes “Einstein’s rest mass–energy equivalence”; artwork uses $E_0 = mc^2$.	Name and rest-energy notation clarified.
B Maxwell’s equations	Four equations in upper background: $\nabla \times B = (1/c)\partial E/\partial t$, $\nabla \times E = -(1/c)\partial B/\partial t$, $\nabla \cdot B = 0$, $\nabla \cdot E = 0$.	Recast in differential-forms notation: $dF = 0$ and $d \star F = \star j$ replace the four vector-calculus equations, in a more compact block.	Notation change: vector calculus $\rightarrow$ differential forms (July 2026 revision).
C Stokes’ theorem	Integral formula in background.	Retained and retypeset.	Typographic cleanup.
D Supergravity	General “Supergravity” background label and formula.	Legend specifies “Minimal supergravity Lagrangian”; formula redrawn next to Dynkin diagram.	Label specificity + redraw.

Item	Original reference	Final version	Change type
E Boundary of a boundary	$\partial\partial = 0$.	$\partial^2 = 0$ in final artwork.	Equivalent notation made more compact.
F Heisenberg uncertainty	$\Delta x \Delta p \geq \hbar/2$.	Retained and retypeset.	Typographic cleanup.
G Riemann zeta function	Euler product/sum relationship.	Retained and retypeset.	Typographic cleanup.
H Feynman diagrams	Particle diagram/string interaction detail.	Legend expanded to “Feynman and string interaction pair-of-pants.”	Label expanded to match content.
I Shannon information theory	No original background entry.	New binary/entropy block added: $H(p) = -\sum p_i \log_2 p_i$ plus binary strings.	New content.
J Bekenstein–Hawking entropy	No original background entry.	New block added at the right of the wall: $S = kAc^3/4G\hbar$, the entropy of a black-hole horizon of area A — gravity, quantum mechanics, thermodynamics, and information in one line.	New content (July 2026 revision).